

Sri Comforts Air Products & Services

Basic Residential Split AC Calculator Logic

This document defines the basic logic for the website AC tonnage calculator using only Length, Width and City.

User Inputs

- Length (ft)
- Width (ft)
- City (current outdoor temperature fetched automatically)

Step 1: Room Area

Area (SFT) = Length × Width

Step 2: Base AC Capacity

Area (SFT)	Base AC
Up to 100	1.0 Ton
101–150	1.5 Ton
151–220	2.0 Ton
221–300	2.5 Ton
301–380	3.0 Ton

Step 3: Temperature Adjustment

Temperature	Adjustment
Below 30°C	No Change
30–35°C	+10%
Above 35°C	+20%

Calculation Logic

1. Calculate Area = Length × Width
2. Determine Base AC from the Area table.
3. Fetch current temperature based on the selected city.
4. If temperature is below 30°C, keep the base AC.
5. If temperature is 30–35°C, increase the base AC by 10%.
6. If temperature is above 35°C, increase the base AC by 20%.
7. Round the result to the nearest available AC size (1.0, 1.5, 2.0, 2.5 or 3.0 Ton).

Example

Length = 10 ft

Width = 12 ft

Area = 120 SFT

Base AC = 1.5 Ton

City = Hyderabad (37°C)

Final AC = $1.5 \times 1.20 = 1.8$ Ton

Recommended AC = 2.0 Ton